

## Model 2

Update: Loss rate  $P$  and delay  $d$

Fixpoint

Init  $P$  &  $d$

TCP  
Model

Avg. Packet rate

Network  
Model

too model can be  
very bad

when result  
stable

- TCP Mechanism and Network behavior mutually dependent.  
To analyze performance model feedback loop
- TCP Model  $\rightarrow$  Closed form throughput formula (very complex)  
Captures avg. behavior of TCP Mechanisms  
(Slow start, cwnd, backoff) (Include slide formula)
- Network model  $\rightarrow$  e.g.  $M/M/1/K$   
Simulates queue bottleneck returns  $P$  &  $d$

## Extra notes

- Model 3 can extend model 2 (Fixpoint) with  
Multiple TCP connections.
- Throttled to service rate  $\frac{S \cdot V}{j}$  when buffer fills.
- Congestion feedback captured by Markov chain  
(no closed form)

$V$  - total service rate

$j$  - # conns

$\delta$  - TCP degradation factor (Do not reach Max TP)